MATH 241: Calculus III

Fall 2026

Lecture 3: Lines and Planes in Space

Background & Motivation

With vectors in hand, we can now describe geometric objects in 3D. A line needs a point and a direction; a plane needs a point and a normal. These are the simplest objects in space, and the techniques here—parametric equations, dot products for angles, cross products for normals—come back throughout the course.

1. Lines in $\mathbb{R}^3$

Definition 1: Parametric Equation of a Line

A line through $P_0 = (x_0, y_0, z_0)$ with direction vector $\mathbf{d} = \langle a, b, c\rangle$:

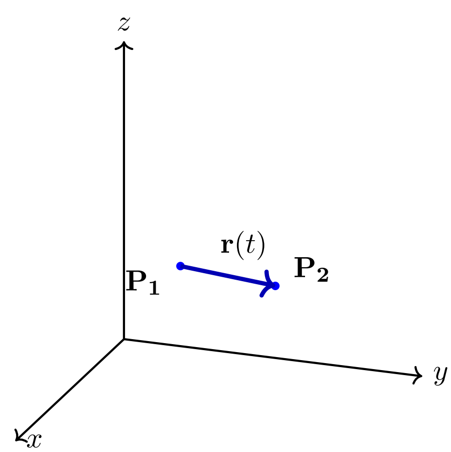
Vector form:

$$\mathbf{r}(t) = \mathbf{r}_0 + t\,\mathbf{d} = \langle x_0 + at,\ y_0 + bt,\ z_0 + ct\rangle$$

Component form:

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

– Each value of $t \in (-\infty, \infty)$ gives a point on the line



The line segment connecting points $\mathbf{P_1}$ and $\mathbf{P_2}$ in 3D, represented by $\mathbf{r}(t) = (1-t)\mathbf{P_1} + t\mathbf{P_2}$ for $t \in [0, 1]$.

Definition 2: Line Segment

The segment from P_1 to P_2:

$$\mathbf{r}(t) = (1-t)\,\mathbf{r}_1 + t\,\mathbf{r}_2, \quad t \in [0, 1]$$

– $t = 0$ gives P_1; $t = 1$ gives P_2

Example 1: Parametric Line

Write the parametric equation of the line through $P_0 = (1, -2, 3)$ in direction $\mathbf{d} = \langle 4, -1, 2 \rangle$.
1. Apply the formula $\mathbf{r}(t) = \mathbf{r}_0 + t\,\mathbf{d}$:

$$\mathbf{r}(t) = \langle 1 + 4t,\ -2 - t,\ 3 + 2t \rangle$$

2. Component form: $x = 1 + 4t, \quad y = -2 - t, \quad z = 3 + 2t$

2. Planes in $\mathbb{R}^3$

Definition 3: Equation of a Plane

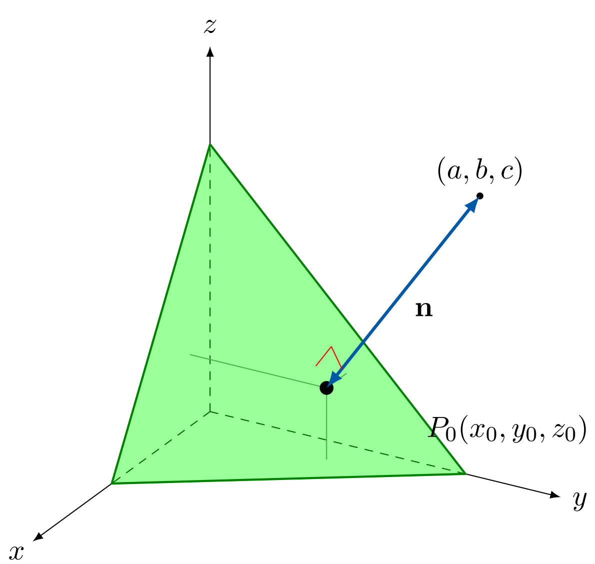
A plane through $P_0 = (x_0, y_0, z_0)$ with normal vector $\mathbf{n} = \langle A, B, C \rangle$:
Point-normal form:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

General form:

$$Ax + By + Cz = D$$

where $D = Ax_0 + By_0 + Cz_0$



A plane in 3D space defined by a point P_0 and a normal vector $\mathbf{n}$ extending from P_0 in the perpendicular direction.

Example 2: Plane from Point and Normal

Find the equation of the plane through $(2, -1, 3)$ with normal $\mathbf{n} = \langle 4, 5, -2 \rangle$.
1. Point-normal form:

$$4(x - 2) + 5(y + 1) - 2(z - 3) = 0$$

2. Expand:

$$4x - 8 + 5y + 5 - 2z + 6 = 0$$

3. Result: $\boxed{4x + 5y - 2z = -3}$

Example 3: Plane from Three Points

Find the equation of the plane through $P_1 = (1, 2, 3)$, $P_2 = (4, -1, 2)$, $P_3 = (-2, 3, 5)$.
1. Two vectors in the plane:

$$\overrightarrow{P_1P_2} = \langle 3, -3, -1 \rangle, \quad \overrightarrow{P_1P_3} = \langle -3, 1, 2 \rangle$$

2. Normal via cross product:

$$\mathbf{n} = \langle 3, -3, -1 \rangle \times \langle -3, 1, 2 \rangle = \langle -6 + 1,\ 3 - 6,\ 3 - 9 \rangle = \langle -5, -3, -6 \rangle$$

3. Use $P_1 = (1, 2, 3)$:

$$-5(x - 1) - 3(y - 2) - 6(z - 3) = 0 \implies \boxed{5x + 3y + 6z = 29}$$

3. Intersections and Angles

Line–Plane Intersection

– Strategy: substitute parametric equations of the line into the plane equation, solve for t

Example 4: Line–Plane Intersection

Find where $\mathbf{r}(t) = \langle 1 + 2t,\ 2 - t,\ 3 + t \rangle$ intersects $3x - y + 2z = 5$.
1. Substitute:

$$3(1 + 2t) - (2 - t) + 2(3 + t) = 5$$

2. Solve:

$$3 + 6t - 2 + t + 6 + 2t = 5 \implies 9t + 7 = 5 \implies t = -\tfrac{2}{9}$$

3. Point:

$$\left(1 - \tfrac{4}{9},\ 2 + \tfrac{2}{9},\ 3 - \tfrac{2}{9}\right) = \boxed{\left(\tfrac{5}{9},\ \tfrac{20}{9},\ \tfrac{25}{9}\right)}$$

Angle Between Two Planes

Angle Between Planes

If planes have normals $\mathbf{n}_1$ and $\mathbf{n}_2$:

$$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{\|\mathbf{n}_1\|\,\|\mathbf{n}_2\|}$$

– Parallel: $\mathbf{n}_1 \parallel \mathbf{n}_2$ – Perpendicular: $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$

Line of Intersection of Two Planes

Strategy

Given planes Π_1 and Π_2 with normals $\mathbf{n}_1$ and $\mathbf{n}_2$:

1. **Direction:** $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$
2. **Point:** Set one variable (e.g. $z = 0$) and solve the 2×2 system

Example 5: Intersection of Two Planes

Find the line of intersection of $2x - y + z = 3$ and $x + y - 2z = -4$.
1. Direction: $\mathbf{n}_1 = \langle 2, -1, 1 \rangle$, $\mathbf{n}_2 = \langle 1, 1, -2 \rangle$

$$\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2 = \langle (-1)(-2) - (1)(1),\ (1)(1) - (2)(-2),\ (2)(1) - (-1)(1) \rangle = \langle 1, 5, 3 \rangle$$

2. Point (set $z = 0$): $2x - y = 3$ and $x + y = -4$.
3. Adding: $3x = -1 \implies x = -1/3$, $y = -11/3$.
4. Result: $\mathbf{r}(t) = \langle -1/3 + t,\ -11/3 + 5t,\ 3t \rangle$

4. Distances

Theorem 1: Distance from a Point to a Plane

Distance from $P_1 = (x_1, y_1, z_1)$ to $Ax + By + Cz = D$:

$$d = \frac{|Ax_1 + By_1 + Cz_1 - D|}{\sqrt{A^2 + B^2 + C^2}}$$

Example 6: Point-to-Plane Distance

Find the distance from $Q = (1, -1, -3)$ to the plane $x + 2y + 3z = 18$.
1. Apply formula:

$$d = \frac{|1 + 2(-1) + 3(-3) - 18|}{\sqrt{1 + 4 + 9}} = \frac{|1 - 2 - 9 - 18|}{\sqrt{14}} = \frac{28}{\sqrt{14}}$$

2. Simplify: $d = \frac{28\sqrt{14}}{14} = \boxed{2\sqrt{14}}$

5. Vector Projection onto a Plane

Projection onto a Plane

To project $\mathbf{a}$ onto a plane with normal $\mathbf{n}$:

$$\mathbf{a}_{\text{plane}} = \mathbf{a} - \operatorname{proj}_{\mathbf{n}} \mathbf{a} = \mathbf{a} - \frac{\mathbf{a} \cdot \mathbf{n}}{\|\mathbf{n}\|^2}\mathbf{n}$$

– Subtract the normal component from the vector

Example 7: Projection onto a Plane

Project $\mathbf{a} = \langle 3, 4, 5\rangle$ onto the plane with normal $\mathbf{n} = \langle 1, 1, 1\rangle$.

1. Normal component:

$$\text{proj}_{\mathbf{n}}\,\mathbf{a} = \frac{3+4+5}{3}\,\langle 1,1,1\rangle = 4\,\langle 1,1,1\rangle = \langle 4,4,4\rangle$$

2. Plane projection:

$$\mathbf{a}_{\text{plane}} = \langle 3,4,5\rangle - \langle 4,4,4\rangle = \langle -1,0,1\rangle$$

3. Check: $\langle -1,0,1\rangle \cdot \langle 1,1,1\rangle = -1+0+1 = 0$ ✓ (perpendicular to $\mathbf{n}$)

Show all work for full credit.

Practice Problems

Problem 1. Write the parametric equation of the line through $P = (2, 1, -3)$ in direction $\mathbf{d} = \langle 1, -2, 4\rangle$.

Answer: $\mathbf{r}(t) = \langle 2+t,\ 1-2t,\ -3+4t\rangle$

Problem 2. Find the equation of the plane through $(3, 0, -1)$ with normal $\mathbf{n} = \langle 2, 1, -5\rangle$.

Answer: $2x + y - 5z = 11$

Problem 3. Find where the line $\mathbf{r}(t) = \langle -t,\ 2t,\ t+3\rangle$ intersects the plane $-x + y = 6$.

Answer: $t = 2, \quad \text{point} = (-2, 4, 5)$

Problem 4. Find the distance from $(2, 3, -1)$ to the plane $x - 2y + 2z = 4$.

Answer: $d = 10/3$

Problem 5. Given $\Pi_1 : x + 2y - z = 3$ and $\Pi_2 : 2x - y + 3z = 1$: (a) find the angle between the planes, (b) find the line of intersection.

Answer (a): $\theta = \arccos\left(\frac{3}{2\sqrt{21}}\right) \approx 70.9^\circ$

Answer (b): $\mathbf{r}(t) = \langle 1 + t,\ 1 - t,\ -t \rangle$